CARDURA XL*
Doxazosin mesylate GITS

1. NAME OF THE MEDICINAL PRODUCT

CARDURA XL

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Active ingredient: doxazosin
The GITS tablets contain doxazosin mesylate equivalent to 4 mg and 8 mg doxazosin.

3. PHARMACEUTICAL FORM

Controlled-release tablets

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Hypertension

Doxazosin GITS is indicated for the treatment of hypertension and can be used as the initial agent to control blood pressure in the majority of patients. In patients not adequately controlled on a single antihypertensive agent, doxazosin may be used in combination with another agent such as a thiazide diuretic, a beta-blocker, a calcium antagonist or an angiotensin-converting enzyme inhibitor.

Benign Prostatic Hyperplasia

Doxazosin GITS is indicated for the treatment of clinical symptoms in benign prostatic hyperplasia (BPH) and for reduced urinary flow associated with BPH. Doxazosin GITS may be used in BPH patients who are either hypertensive or normotensive. While the blood pressure changes in normotensive patients with BPH are clinically insignificant, patients with hypertension and BPH have had both the conditions effectively treated with doxazosin GITS monotherapy.

4.2 Posology and Method of Administration

Hypertension and Benign Prostatic Hyperplasia

Doxazosin GITS can be taken with or without food.

The GITS tablets should be swallowed whole with a sufficient amount of liquid. Patients should not chew, divide or crush the tablets (see section **4.4 Special Warnings and Precautions for Use**: *Information for Patients*).

The initial dose of doxazosin GITS is 4 mg once daily. Over 50% of patients with mild to moderate severity hypertension will be controlled on doxazosin GITS 4 mg once daily. The optimal effect of doxazosin may take up to 4 weeks. If necessary, the dosage may be increased following this period to 8 mg once daily according to patient response.

The maximum recommended dose is 8 mg once daily.

Use in Elderly

Normal adult dosage is recommended.

Use in Renally Impaired Patients

Doxazosin is not dialysable. Since the pharmacokinetics of doxazosin are unchanged in patients with renal insufficiency, and there is no evidence that doxazosin aggravates existing renal dysfunction, the usual dosages may be used in these patients.

Use in Hepatically Impaired Patients

See section **4.4. Special Warnings and Precautions for Use**.

Use in Children

The safety and efficacy of doxazosin in children have not been established.

4.3 Contraindications

Doxazosin GITS is contraindicated in patients with a known hypersensitivity to quinazolines, doxazosin, or any of the inert ingredients.

4.4 Special Warnings and Precautions for Use

Postural Hypotension/Syncope

As with all alpha-blockers, a very small percentage of patients have experienced postural hypotension evidenced by dizziness and weakness, or rarely loss of consciousness (syncope), particularly with the commencement of therapy. When instituting therapy with any effective alpha-blocker, the patient should be advised how to avoid symptoms resulting from postural hypotension and what measures to take should they develop. The patient should be cautioned to avoid situations where injury could result should dizziness or weakness occur during the initiation of doxazosin therapy.

Use with Phosphodiesterase Type-5 Inhibitors

Concomitant administration of doxazosin with a phosphodiesterase type-5 (PDE-5) inhibitor should be used with caution as it may lead to symptomatic hypotension in some patients. No studies have been conducted with doxazosin GITS.

Impaired Hepatic Function

As with any drug wholly metabolized by the liver, doxazosin should be administered with caution to patients with evidence of impaired hepatic function (see section **5.2 Pharmacokinetic Properties**).

Gastrointestinal Disorders

Markedly reduced gastrointestinal (GI) retention times of doxazosin GITS may influence the pharmacokinetic profile and hence the clinical efficacy of the drug. As with any other non-deformable material, caution should be used when administering doxazosin GITS in patients with preexisting severe GI narrowing (pathologic or iatrogenic).

There have been rare reports of obstructive symptoms in patients with known strictures in association with the ingestion of another drug in this non-deformable sustained-release formulation.

Intraoperative Floppy Iris Syndrome

The intraoperative floppy iris syndrome (IFIS), a variant of small pupil syndrome, has been observed during cataract surgery in some patients on or previously treated with alpha-1-blockers. As IFIS may lead to increased procedural complications during the operation, current or past use of alpha-blockers should be made known to the ophthalmologic surgeon in advance of surgery.

Priapism

Prolonged erections and priapism have been reported with alpha-1 blockers including doxazosin in post marketing experience. In the event of an erection that persists longer than 4 hours, the patient should seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency could result.

Information for Patients

Patients should be informed that doxazosin GITS should be swallowed whole. Patients should not chew, divide or crush the tablets. Patients should not be concerned if they occasionally notice in their stool something that looks like a tablet. In doxazosin GITS, the medication is contained within a non-absorbable shell that has been specially designed to slowly release the drug so the body can absorb it. When this process is completed, the empty tablet is eliminated from the body.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Use with PDE-5 Inhibitors

See section **4.4 Special Warnings and Precautions for Use:** *Use with Phosphodiesterase Type-5 Inhibitors*.

CYP3A4 Inhibitors

In vitro studies suggest that doxazosin is a substrate of CYP 3A4. Caution should be exercised when concomitantly administering doxazosin with a strong CYP3A4 inhibitor, such as clarithromycin, indinavir, itraconazole, ketoconazole, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin or voriconazole (see section **5.2 Pharmacokinetic properties**).

Other

Most (98%) of the plasma doxazosin is protein bound. *In vitro* data in human plasma indicate that doxazosin has no effect on protein binding of digoxin, warfarin, phenytoin or indomethacin. Doxazosin has been administered without any adverse drug interaction in clinical experience with thiazide diuretics, frusemide, beta-blockers, non-steroidal anti-inflammatory drugs, antibiotics, oral hypoglycemic drugs, uricosuric agents, or anticoagulants.

4.6 Fertility, Pregnancy and Lactation

Although no teratogenic effects were seen in animal testing with doxazosin, reduced fetal survival was observed in animals at extremely high doses. These doses were approximately 300 times the maximum human recommended dose.

A single case report demonstrated transfer of doxazosin into human breast milk and animal studies have shown that doxazosin accumulates in breast milk (see section **5.3 Preclinical safety data**).

As there are no adequate and well-controlled studies in pregnant or nursing women, the safety of doxazosin GITS during pregnancy or lactation has not yet been established. Accordingly, during pregnancy or lactation, doxazosin GITS should be used only when, in the opinion of the physician, the potential benefit outweighs the potential risk.

4.7 Effects on Ability to Drive and Use Machines

The ability to engage in activities such as operating machinery or operating a motor vehicle may be impaired, especially when initiating doxazosin GITS therapy.

4.8 Undesirable Effects

The following lists the common (>1%) adverse events reported in pre-marketing placebo-controlled clinical trials with doxazosin GITS. It is important to emphasize that events reported during therapy may not necessarily be caused by the therapy.

Hypertension

Cardiac Disorder: palpitation, tachycardia

Ear and Labyrinth Disorders: vertigo

Gastrointestinal Disorders: abdominal pain, mouth dry, nausea

General Disorders and Administration Site Conditions: asthenia, chest pain, peripheral edema

Musculoskeletal and Connective Tissue Disorders: back pain, myalgia

Vascular Disorders: postural hypotension

Nervous System Disorders: dizziness, headache

Respiratory, Thoracic and Mediastinal Disorders: bronchitis, coughing

Skin and Subcutaneous Tissue Disorders: pruritus

Renal and Urinary Disorders: cystitis, urinary incontinence

Benign Prostatic Hyperplasia

Ear and Labyrinth Disorders: vertigo

General Disorders and Administration Site Conditions: asthenia, peripheral edema

Gastrointestinal Disorders: abdominal pain, dyspepsia, nausea

Infection and Infestations: influenza-like symptoms, respiratory tract infection, urinary tract infection

Musculoskeletal and Connective Tissue Disorders: back pain, myalgia

Nervous System Disorders: dizziness, headache, somnolence

Respiratory, Thoracic and Mediastinal Disorders: bronchitis, dyspnea, rhinitis

Vascular Disorders: hypotension, postural hypotension

The incidence of adverse events following treatment with doxazosin GITS (41%) in clinical studies of patients with BPH was broadly similar to that following placebo (39%) and less than that following standard doxazosin (54%).

The adverse event profile in elderly (>65 years) BPH patients showed no difference from the profile in the younger population.

In post-marketing experience, the following additional adverse events have been reported:

Blood and Lymphatic Disorders: leukopenia, thrombocytopenia

Ear and Labyrinth Disorders: tinnitus

Eye Disorders: blurred vision, IFIS (see section **4.4 Special Warnings and Precautions for Use**).

Gastrointestinal Disorders: gastrointestinal obstruction, constipation, diarrhea, dyspepsia, flatulence, dry mouth, vomiting

General Disorders and Administration Site Conditions: fatigue, malaise, pain

Hepatobiliary Disorders: cholestasis, hepatitis, jaundice

Immune System Disorders: allergic reaction

Investigations: abnormal liver function tests, weight increase

Metabolism and Nutrition: anorexia

Musculoskeletal and Connective Tissue Disorders: arthralgia, muscle cramps, muscle weakness

Nervous System Disorder: dizziness postural, hypoesthesia, paresthesia, syncope, tremor

Psychiatric Disorders: agitation, anxiety, depression, insomnia, nervousness

Renal and Urinary Disorders: dysuria, hematuria, micturition disorder, micturition frequency, nocturia, polyuria, urinary incontinence

Reproductive System and Breast Disorder: gynecomastia, impotence, priapism, retrograde ejaculation

Respiratory, Thoracic and Mediastinal Disorders: bronchospasm aggravated, coughing, dyspnea, epistaxis

Skin/Appendages: alopecia, pruritus, purpura, skin rash, urticaria

Vascular Disorders: hot flushes, hypotension

The following additional adverse events have been reported in marketing experience among patients treated for hypertension but these, in general, are not distinguishable from symptoms that might have occurred in the absence of exposure to doxazosin: bradycardia, tachycardia, palpitation, chest pain, angina pectoris, myocardial infarction, cerebrovascular accidents, and cardiac arrhythmias.

4.9 Overdose

Should overdosage lead to hypotension, the patient should be immediately placed in a supine, head-down position. Other supportive measures should be performed if thought appropriate in individual cases. Since doxazosin is highly protein bound, dialysis is not indicated.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Hypertension

Administration of doxazosin GITS to hypertensive patients causes a clinically significant reduction in blood pressure as a result of a reduction in systemic vascular resistance. This effect is thought to result from selective blockade of the alpha-1-adrenoceptors located in the vasculature. With once-daily dosing, clinically significant reductions in blood pressure are present throughout the day and at 24 hours post dose. The majority of patients are controlled on the initial 4 mg dose of doxazosin GITS. In patients with hypertension, blood pressure reductions during treatment with doxazosin GITS were similar in both the sitting and standing positions.

Subjects treated with standard doxazosin for hypertension can be transferred to doxazosin GITS and titrated upwards, as needed, while maintaining efficacy and tolerability.

Unlike non-selective alpha-adrenoceptor blocking agents, tolerance has not been observed in long-term therapy with doxazosin GITS. Elevations of plasma renin activity and tachycardia were seen infrequently in sustained doxazosin therapy.

Doxazosin produces favorable effects on blood lipids, with a significant increase in the high-density lipoprotein (HDL)/total cholesterol ratio and significant reductions in total triglycerides and total cholesterol. It, therefore, confers an advantage over diuretics and beta-adrenoceptor blocking agents, which adversely affect these parameters. Based on the established association of hypertension and blood lipids with coronary heart disease, the favorable effects of doxazosin therapy on both blood pressure and lipids indicate a reduction in risk of developing coronary heart disease.

Treatment with doxazosin has been shown to result in regression of left ventricular hypertrophy, inhibition of platelet aggregation, and enhanced tissue plasminogen activator capacity. Additionally, doxazosin improves insulin sensitivity in patients who have impairment.

Doxazosin has been shown to be free of adverse metabolic effects and is suitable for use in patients with asthma, diabetes, left ventricular dysfunction, and gout.

An *in vitro* study has demonstrated the antioxidant properties of the 6'- and 7'-hydroxy metabolites of doxazosin at concentrations of 5 micromolar.

Benign Prostatic Hyperplasia

Administration of doxazosin GITS to patients with symptomatic BPH results in a significant improvement in urodynamics and symptoms. The effect in BPH is thought to result from selective blockade of the alpha-adrenoceptors located in the muscular stroma and capsule of the prostate and in the bladder neck.

Doxazosin has been shown to be an effective blocker of the 1A subtype of the alpha-1-adrenoceptor, which accounts for over 70% of the subtypes in the prostate. This accounts for the action in BPH patients.

Doxazosin GITS has demonstrated sustained efficacy and safety in the long term-treatment of BPH.

Doxazosin GITS given in the recommended dosage regimen has little or no effect on blood pressure in normotensive patients.

In a controlled clinical BPH trial, treatment with doxazosin in patients with sexual dysfunction was associated with improvement in sexual function.

5.2 Pharmacokinetic Properties

Absorption

After oral administration of therapeutic doses, doxazosin GITS is well absorbed with peak blood levels gradually reached at 8 to 9 hours after dosing. Peak plasma levels are approximately one-third of those of the same dose of standard doxazosin tablets. Trough levels at 24 hours are, however, similar.

The pharmacokinetic characteristics of doxazosin GITS will lead to a smoother plasma profile.

Peak/trough ratio of doxazosin GITS is less than half that of standard doxazosin tablets.

At steady state, the relative bioavailability of doxazosin from doxazosin GITS compared to the standard form was 54% at the 4 mg dose and 59% at the 8 mg dose.

Pharmacokinetic studies with doxazosin GITS in the elderly have shown no significant alterations compared to younger patients.

Biotransformation/Elimination

The plasma elimination is biphasic, with the terminal elimination half-life being 22 hours. This provides the basis for once-daily dosing. Doxazosin is extensively metabolized, with <5% excreted as unchanged drug.

Pharmacokinetic studies with standard doxazosin in patients with renal impairment have shown no significant alterations compared to patients with normal renal function.

There are only limited data in patients with liver impairment and on the effects of drugs known to influence hepatic metabolism (e.g., cimetidine). In a clinical study in 12 subjects with moderate hepatic impairment, single-dose administration of doxazosin resulted in an increase in AUC of 43% and a decrease in apparent oral clearance of 40%. As with any drug wholly metabolized by the liver, use of doxazosin in patients with altered liver function should be undertaken with caution (see section **4.4 Special Warnings and Precautions for Use**).

Approximately 98% of doxazosin is protein-bound in plasma.

Doxazosin is primarily metabolized by O-demethylation and hydroxylation.

Doxazosin is extensively metabolized in the liver. In vitro studies suggest that the primary pathway for elimination is via CYP 3A4; however, CYP 2D6 and CYP 2C9 metabolic pathways are also involved for elimination, but to a lesser extent.

5.3 Preclinical Safety Data

Carcinogenesis

Chronic dietary administration (up to 24 months) of doxazosin at maximally tolerated doses of 40 mg/kg/day in rats and 120 mg/kg/day in mice revealed no evidence of carcinogenic potential. The highest doses evaluated in the rat and mouse studies are associated with AUCs (a measure of systemic exposure) that are 8 times and 4 times the human AUC at a dose of 16 mg/day, respectively.

Mutagenesis

Mutagenicity studies revealed no drug- or metabolite-related effects at either chromosomal or subchromosomal levels.

Impairment of Fertility

Studies in rats showed reduced fertility in males treated with doxazosin at oral doses of 20 mg/kg/day (but not 5 or 10 mg/kg/day), about 4 times the human AUC at a dose of 12 mg/day. This effect was reversible within 2 weeks of drug withdrawal. There have been no reports of any effects of doxazosin on male fertility in humans.

Lactation

Studies in lactating rats given a single oral dose of 1 mg/kg of [2-^{14}C]-doxazosin indicate that doxazosin accumulates in rat breast milk with a maximum of concentration about 20 times greater than the maternal plasma concentration.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Doxazosin mesylate GITS tablets include the following inert ingredients: polyethylene oxide, sodium chloride, hydroxypropyl methylcellulose, red ferric oxide (E172), titanium dioxide (E171), magnesium stearate, cellulose acetate, macrogol, pharmaceutical glaze and black iron oxide (E172).

6.2 Incompatibilities

None

6.3 Shelf-life

Please refer to outer carton.

6.4 Special Precautions for Storage

Store below 30°C

6.5 Nature and Contents of Container

Doxazosin Mesylate GITS 4 mg and 8 mg modified release tablets are available in blister-pack of 30’s and 100’s.

Some product strengths or pack sizes may not be available in your country.

6.6 Special Instructions for Use/Handling

Not Applicable.

6.7 Manufacturer

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United States

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